Editorial for the Test Contest

A. Sum of Numbers I

The answer is $n + m$: it suffices to read the two numbers, add them and print the result. Under the constraint $(1 \le n, m \le 1000)$ the sum does not exceed 2000 and fits in any integer type, so overflow is impossible. The running time is $O(1)$. ■

B. Sum of Numbers II

The solution is the same as in problem A. The subgroups differ only in their constraints: $1 \le n, m \le 500$ in the first one and up to 1000 in the second. The algorithm does not change, so no separate code for the first subgroup is needed: a correct program passes both subgroups and receives 100 points. ■

C. Sum of Numbers III

The answer is $n + m$, as in problems A and B. Full marks for subgroup 1 are awarded for passing all of its tests, which include the sample tests. The problem has no other peculiarities. ■

D. Sum of Numbers IV

The algorithm is the same: the answer is $n + m$. The distinctive feature is the merged-subgroup scoring: for each subgroup the best result among all submissions is taken separately, and the points for the problem equal the sum of these maxima. Thus a submission that passes only subgroup 1 and another one that passes only subgroup 2 together bring the full 100 points. A correct solution passes all the tests of both subgroups and receives 100 points with a single submission. ■

E. Sum of Numbers V

The answer is $n + m$. The difference from problems A–D lies in the scoring system: the two tests besides the sample tests are scored independently, 50 points each, rather than as a single subgroup. This does not affect the solution: the problem admits no partial solution, it is either solved completely or not solved at all. ■

F. Guess the Number

Binary search applies. Keep the bounds of the interval of possible values $[\mathrm{lo}, \mathrm{hi}]$, initially $\mathrm{lo} = 1$, $\mathrm{hi} = t$. At each step make the query $x = \left\lfloor \frac{\mathrm{lo} + \mathrm{hi}}{2} \right\rfloor$:

- on the reply = the number has been guessed, and the program must terminate immediately;
- on the reply < the sought number is less than x, and the interval shrinks to $[\mathrm{lo}, x - 1]$;
- on the reply > the interval shrinks to $[x + 1, \mathrm{hi}]$.

The length of the interval decreases by at least half with every query, so the number of queries does not exceed $\lceil \log_2 t \rceil$; for $t \le 10000$, 14 queries suffice. The output buffer must be flushed after every query, and the program must terminate immediately after receiving the reply =. ■

G. Add 1

A run-twice problem is solved in parts. In the first run the program reads the number n and prints $n + 1$. In the second run its input is the value printed by the first run: the value is transferred by the system, so no temporary files are needed. The program must print $v - 1$, that is, recover the original n.

The main difficulty of the problem is technical rather than algorithmic: both runs must use only the standard input and output streams, print nothing extraneous and terminate correctly. ■